

The Monumental Vote on N

A Unified Theory of Discount-Factor Elimination, Economic Constitutional Choice, Political Mastery, Moral Deliberation, and Republican Self-Examination

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Abstract

This paper develops a formal and interdisciplinary reconstruction of a constitutional-economic deliberation concerning the choice between one and two post-elimination economies. The underlying theoretical architecture contains multiple methods for eliminating an exponential discounter. One class of elimination produces a single economy characterized by a zero risk-free rate and zero risk premium. Another admissible construction produces two economies, one characterized by $(r_f, p) = (0, 0)$ and the other by $(r_f, p) = (1, 0)$.

The resulting choice is therefore not merely mathematical. It becomes a problem in economics, finance, political science, philosophy, psychology, welfare economics, international relations, institutional design, computer science, algorithmic decision-making, and moral theory. The initial deliberation produces a 4–3 majority in favour of $N = 1$. Rather than treating the majority as sufficient, the Republic examines the specific reasons for dissent. The minority is found not to possess a unified intention: one vote defends principled plurality, another distrusts concentrated unity, and a third seeks comparative resilience.

After the full information set is incorporated into deliberation, each dissenting voter independently revises the earlier position. The final outcome is therefore not unanimity by suppression but unanimity through dissent, examination, information, reconsideration, and independent judgment.

The paper proposes a general constitutional principle: a sufficiently self-critical republic must examine not only its rulers and institutions, but also the information sets, feasible sets, selection mechanisms, discounting mechanisms, and higher-order architectures that determine the conditions under which political and economic choices are made.

The paper ends with “The End”

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1 Introduction

A political community may vote on a number without understanding what the number represents. Conversely, a mathematical system may admit several solutions without requiring that every admissible solution be retained as an economic institution.

The present paper studies a choice between

$$N = 1$$

and

$$N = 2.$$

The choice emerges after the elimination of an exponential discounter through more than one theoretical route. The central issue is therefore not simply whether the exponential discounter should be retained. The relevant problem is the structure of the economy after elimination.

The first architecture produces

$$N = 1 \implies (r_f, p) = (0, 0),$$

where r_f denotes the risk-free rate and p denotes the risk premium.

The alternative architecture produces

$$N = 2 \implies \{(r_f, p) = (0, 0), (r_f, p) = (1, 0)\}.$$

Hence the decision can be written as

$$\mathcal{C} = \{N = 1, N = 2\}. \tag{1}$$

However, the numerical distinction conceals a deeper constitutional difference. The choice is between a single economy satisfying

$$r_f = 0, \quad p = 0,$$

and a two-economy structure containing, in addition, an economy with

$$r_f = 1 = 100\%, \quad p = 0.$$

The central research question is therefore:

How should a Republic choose among mathematically admissible economic architectures when those architectures differ in financial, political, psychological, moral, and institutional consequences?

The answer developed here is not a simple majoritarian rule. Instead, the decision procedure is

Dissent $\rightarrow$ Testimony $\rightarrow$ Examination $\rightarrow$ Full Information $\rightarrow$ Reconsideration $\rightarrow$ Unanimity.

2 The Discount-Factor Problem

2.1 Basic notation

Let

$$r_f \geq 0$$

denote the risk-free rate and let

$$p \geq 0$$

denote the risk premium.

A conventional discount factor may be represented as

$$D(r) = \frac{1}{1+r}. \quad (2)$$

More generally, if a stream of future value is discounted over t periods, an exponential discounting mechanism takes the form

$$D_t = (1+r)^{-t}. \quad (3)$$

The theoretical question considered here is whether the exponential discounter should remain part of the economic architecture when alternative elimination mechanisms produce economies with zero risk-free rate and zero risk premium.

2.2 The single-economy outcome

One elimination route yields the unique post-elimination condition

$$r_f = 0, \quad p = 0. \quad (4)$$

Thus,

$$N = 1. \quad (5)$$

The economic interpretation is immediate:

$$D = \frac{1}{1+r_f} = 1. \quad (6)$$

There is no positive risk-free return generated by the selected structure, and the risk premium is also zero.

2.3 The two-economy outcome

An alternative construction yields

$$(r_f, p) = (0, 0) \quad (7)$$

or

$$(r_f, p) = (1, 0). \quad (8)$$

Hence

$$N = 2. \quad (9)$$

The second economy therefore contains

$$r_f = 1,$$

which corresponds to

$$100\%$$

in rate notation.

Consequently,

$$D(1) = \frac{1}{1+1} = \frac{1}{2}. \quad (10)$$

The central constitutional question is therefore not whether two mathematical solutions exist. They do. The question is whether the Republic should preserve both as part of its economic architecture.

3 Formal Choice Architecture

Define the admissible post-elimination set as

$$\Omega = \{\omega_1, \omega_2\}, \quad (11)$$

where

$$\omega_1 = (r_f, p) = (0, 0), \quad (12)$$

$$\omega_2 = (r_f, p) = (1, 0). \quad (13)$$

The institutional alternatives are therefore

$$\mathcal{E}_1 = \{\omega_1\}, \quad (14)$$

$$\mathcal{E}_2 = \{\omega_1, \omega_2\}. \quad (15)$$

The numerical problem becomes

$$N = |\mathcal{E}|. \quad (16)$$

Thus,

$$N \in \{1, 2\}. \quad (17)$$

This formulation distinguishes an important pair of concepts:

Mathematical admissibility $\neq$ Institutional desirability.

The existence of ω_2 establishes that ω_2 is an admissible solution under the stated theorem architecture. It does not, by itself, establish that the Republic should preserve ω_2 .

4 The Initial Vote

The first deliberation produces the following positions.

Table 1: Initial vote on the number of economies

Voter	Vote
Plato	$N = 1$
Kant	$N = 2$
Descartes	$N = 1$
Schopenhauer	$N = 2$
Machiavelli	$N = 2$
Soumadeep Ghosh	$N = 1$
Sisyphus	$N = 1$

Hence,

$$V_1 = 4, \tag{18}$$

$$V_2 = 3. \tag{19}$$

The majority rule yields

$$N^* = 1. \tag{20}$$

However, this paper rejects the proposition that

$$4 > 3$$

is sufficient to terminate inquiry.

A majority identifies a provisional decision. It does not necessarily identify the full structure of disagreement.

5 The Theory of Minority Examination

5.1 Definition

Definition 1 (Minority Examination). *Given a collective choice with majority outcome x^* and dissenting set $\mathcal{D}$, a minority examination is a procedure that identifies the specific reasons, information sets, intentions, and assumptions underlying each dissenting vote before the decision is treated as epistemically complete.*

Let voter i possess

$$\mathcal{I}_i = (A_i, B_i, M_i, R_i), \tag{21}$$

where

- A_i is the set of admissible alternatives perceived by voter i ;
- B_i is the belief structure of voter i ;
- M_i is the motivational or normative structure of voter i ;
- R_i is the explicit reason for the vote.

Two voters may cast the same ballot while possessing entirely different structures:

$$v_i = v_j \not\Rightarrow \mathcal{I}_i = \mathcal{I}_j. \quad (22)$$

This principle is decisive in the present case.

5.2 The three dissenting intentions

The three $N = 2$ votes are not equivalent.

Table 2: Structure of the initial dissent

Voter	Immediate Reason	Underlying Intention
Kant	Preserve an admissible distinction unless elimination is justified	Prevent arbitrary elimination and protect principled critical reason
Schopenhauer	Do not confuse unity with liberation	Prevent concentration, domination, and false identification of unity with freedom
Machiavelli	Two structures permit comparison	Maintain comparative resilience and detect hidden institutional weakness

Therefore,

$$\boxed{N = 2 \neq \text{a unified intention.}} \quad (23)$$

6 Kant's Reconsideration

Kant's initial argument can be formalized as follows.

Suppose that

$$|\Omega| = 2. \quad (24)$$

Then the existence of two admissible solutions does not automatically imply that one may be deleted merely because another is preferred.

Thus,

$$|\Omega| = 2 \not\Rightarrow N = 1. \quad (25)$$

However, the converse does not follow either:

$$|\Omega| = 2 \not\Rightarrow N = 2 \text{ must be institutionally preserved.} \quad (26)$$

Kant's critical principle therefore requires a justification function

$$J(\omega_k). \quad (27)$$

The Republic may eliminate ω_k if a sufficient public justification exists:

$$J(\omega_k) \geq J^*. \quad (28)$$

The full specification of the second economy changes the information set from an abstract plurality to

$$\omega_2 = (1, 0). \quad (29)$$

Kant's reconsidered logic is therefore

Plurality is not sacred merely because it is plurality.

Hence,

$$\boxed{\text{Two admissible solutions} \not\Rightarrow \text{Two retained institutions.}} \quad (30)$$

Kant revises the vote to

$$v_K = 1. \quad (31)$$

7 Schopenhauer's Reconsideration

Schopenhauer's initial concern is not the positive endorsement of the second economy. It is the possibility that

$$N = 1 \rightarrow \text{concentration} \rightarrow \text{loss of alternatives} \rightarrow \text{domination}. \quad (32)$$

The argument is structurally precautionary.

However, after the economic content of the second alternative becomes explicit, Schopenhauer confronts the reciprocal question:

$$\text{What does plurality preserve?} \quad (33)$$

The existence of an alternative does not imply that the alternative is beneficial:

$$\boxed{\text{Alternative} \neq \text{Virtue.}} \quad (34)$$

Thus, if the principal reason for preserving ω_2 is simply that ω_2 exists, then the argument lacks an independent welfare or moral criterion.

Schopenhauer therefore revises the vote:

$$v_S = 1. \quad (35)$$

8 Machiavelli's Reconsideration

Machiavelli's initial argument is epistemic and political.

Two structures permit comparison:

$$N = 2 \rightarrow \text{comparison} \rightarrow \text{detection of weakness}. \quad (36)$$

This argument is consistent with a theory of Political Mastery in which visible political action is only part of the relevant causal structure.

Let

$$M \rightarrow A \rightarrow C, \quad (37)$$

where

- M denotes a Master or higher-order mastery mechanism;
- A denotes Political Actors;

- C denotes Political Citizens.

The essential insight is that M need not be a particular person. It may represent a mechanism that shapes the conditions under which action occurs.

Let the actor's feasible set be

$$A_i = \{a_{i1}, a_{i2}, \dots, a_{in}\}. \quad (38)$$

The deeper political question is then:

$$\boxed{\text{Who determines } A_i?} \quad (39)$$

Similarly, define

$$I_i = \text{information set}, \quad (40)$$

$$U_i = \text{incentive or utility structure}, \quad (41)$$

$$E_i = \text{expectation structure}, \quad (42)$$

$$S_i = \text{institutional structure}. \quad (43)$$

Political mastery can therefore operate through

$$M : (I_i, U_i, E_i, S_i, A_i) \mapsto (I'_i, U'_i, E'_i, S'_i, A'_i). \quad (44)$$

Machiavelli's revised insight is that comparison need not require permanent institutional retention of every alternative.

The Republic may compare

- counterfactual economies;
- historical economies;
- simulated economies;
- theoretical alternatives;
- out-of-sample scenarios;
- stress-tested institutional designs.

Therefore,

$$\boxed{\text{Comparative method} \neq \text{Permanent institutional plurality}.} \quad (45)$$

Machiavelli formally revises the vote:

$$v_M = 1. \quad (46)$$

9 Sisyphus and the Testimony Against Unnecessary Repetition

Sisyphus enters the analysis as a philosophical witness concerning repetition, recurrence, and the institutionalization of unnecessary burdens.

The symbolic interpretation is:

$$\text{Second economy} \rightarrow \text{second structure} \rightarrow \text{additional institutional burden.} \quad (47)$$

Sisyphus does not prove that every second economy is undesirable. His testimony instead introduces a principle of unnecessary repetition:

$$\boxed{\text{The existence of a second path does not make it worth taking.}} \quad (48)$$

Let $B(N)$ denote institutional burden. Then a simple parsimony criterion is

$$B(2) > B(1) \quad (49)$$

unless the second economy supplies a compensating benefit

$$\Delta W_2 > 0, \quad (50)$$

where W denotes social welfare.

If no such compensating welfare improvement is demonstrated, institutional parsimony supports

$$N = 1. \quad (51)$$

Sisyphus therefore votes

$$v_{Sy} = 1. \quad (52)$$

10 The Final Vote

After minority examination and reconsideration, the final voting matrix is

Table 3: Final vote under the completed information set

Voter	Final Vote
Plato	$N = 1$
Kant	$N = 1$
Descartes	$N = 1$
Schopenhauer	$N = 1$
Machiavelli	$N = 1$
Soumadeep Ghosh	$N = 1$
Sisyphus	$N = 1$

Thus,

$$V_1 = 7, \quad (53)$$

$$V_2 = 0. \quad (54)$$

Hence,

$$\boxed{N^* = 1.} \quad (55)$$

The selected economy is

$$\boxed{r_f = 0, \quad p = 0.} \quad (56)$$

11 A Logical Proof of Deliberative Unanimity

Theorem 2 (Unanimity Through Examination). *Suppose an initial collective choice produces a majority outcome with non-empty dissenting set $\mathcal{D}$. Suppose further that:*

- (i) *each dissenting voter is permitted to state an independent reason;*
- (ii) *the reasons are examined rather than aggregated into a single artificial intention;*
- (iii) *relevant information missing from the original decision is added to the information set;*
- (iv) *each voter retains the freedom to reconsider independently;*
- (v) *no voter is compelled to revise.*

If every dissenting voter independently changes the vote after the completed information set is considered, then the resulting unanimity is deliberative rather than suppressive.

Proof. Let the initial vote vector be

$$V^{(0)} = (v_1^{(0)}, \dots, v_n^{(0)}). \quad (57)$$

Let $\mathcal{D}$ be the set of voters whose initial votes differ from the majority outcome x^* . For every $i \in \mathcal{D}$, observe the reason vector

$$R_i = (A_i, B_i, M_i, R_i). \quad (58)$$

The examination procedure transforms the information structure into

$$\mathcal{I}_i^{(1)} = \mathcal{I}_i^{(0)} \cup \mathcal{F}, \quad (59)$$

where $\mathcal{F}$ is the relevant additional information.

Each voter then independently computes a revised vote

$$v_i^{(1)} = \phi_i(\mathcal{I}_i^{(1)}). \quad (60)$$

If

$$v_i^{(1)} = x^* \quad (61)$$

for all $i \in \mathcal{D}$, and no coercive transformation of the voting function ϕ_i occurs, then unanimity arises from independent revision under a completed information set.

Therefore,

$$\boxed{\text{Dissent} \rightarrow \text{Examination} \rightarrow \text{Information} \rightarrow \text{Independent Revision} \rightarrow \text{Unanimity}.} \quad (62)$$

The conclusion follows. $\square$

12 Economics and Welfare Economics

Let social welfare under architecture N be

$$W(N) = B(N) - C(N) - R(N), \quad (63)$$

where

- $B(N)$ denotes aggregate benefits;
- $C(N)$ denotes institutional and coordination costs;
- $R(N)$ denotes systemic or moral risk.

A welfare criterion favouring $N = 1$ is

$$W(1) > W(2). \quad (64)$$

The point of the deliberation is not that this inequality follows automatically from arithmetic. Rather, the Republic introduces additional normative criteria.

Define

$$\Psi(N) = \alpha W(N) + \beta M(N) + \gamma A(N) + \delta P(N), \quad (65)$$

where

- $M(N)$ denotes moral acceptability;
- $A(N)$ denotes autonomy and institutional self-examination;
- $P(N)$ denotes parsimony;
- $\alpha, \beta, \gamma, \delta \geq 0$ are constitutional weights.

The collective choice is

$$N^* = \arg \max_{N \in \{1,2\}} \Psi(N). \quad (66)$$

This transforms the problem from a narrow rate comparison into a constitutional welfare problem.

13 Finance and the Temporal Structure of the Republic

Discount factors determine how present institutions value future claims.

Let

$$PV = \sum_{t=0}^T D_t X_t. \quad (67)$$

Under exponential discounting,

$$D_t = (1 + r)^{-t}. \quad (68)$$

The present paper considers the possibility that the exponential discounter is itself an object of institutional choice.

This produces a higher-order question:

$$\boxed{\text{Who determines the function by which the future is valued?}} \quad (69)$$

The problem therefore moves beyond asset pricing.

It becomes a problem of constitutional time.

A Republic may elect its rulers and still fail to examine the discounting mechanism through which future generations are represented.

14 Physics: State Space, Stability, and Phase Selection

The two economic architectures may be represented as states of a system.

Let

$$x = (r_f, p). \quad (70)$$

The state space is

$$\mathcal{X} = \{(0, 0), (1, 0)\}. \quad (71)$$

The one-economy regime selects the state

$$x_1 = (0, 0). \quad (72)$$

The two-economy regime retains

$$\mathcal{X}_2 = \{x_1, x_2\}, \quad (73)$$

where

$$x_2 = (1, 0). \quad (74)$$

A physical analogy may be constructed through an energy functional

$$\mathcal{H}(x). \quad (75)$$

The Republic's choice can then be interpreted as selecting a stable constitutional-economic phase

$$x^* = \arg \min_{x \in \mathcal{X}} \mathcal{H}(x), \quad (76)$$

subject to moral, institutional, and welfare constraints.

The analogy is conceptual rather than a claim that the economy literally obeys thermodynamic laws.

15 Chemistry: Reaction Paths and Elimination Mechanisms

Multiple elimination mechanisms may be understood by analogy with alternative reaction pathways.

Suppose an initial economic architecture is represented by

$$E_0. \quad (77)$$

Different elimination procedures are

$$E_0 \xrightarrow{\mathcal{R}_1} E_1 \quad (78)$$

and



The existence of multiple reaction paths does not imply that all products must be retained in the final institutional system.

The relevant constitutional question is

$$\boxed{\text{Which resulting state should be stabilized?}} \quad (80)$$

This is analogous to product selection under competing reaction pathways, although the analogy does not imply literal chemical equivalence.

16 Biology and Evolutionary Selection

An institutional architecture may also be represented as a population of economic states.

Let the population be

$$\mathcal{P} = \{\omega_1, \omega_2, \dots, \omega_N\}. \quad (81)$$

A selection mechanism assigns fitness

$$F(\omega_i). \quad (82)$$

However, constitutional selection differs from biological natural selection because the Republic can explicitly alter the fitness criterion.

Thus,

$$F = F_{\text{economic}} + F_{\text{moral}} + F_{\text{institutional}}. \quad (83)$$

This distinction is crucial. Human institutions can choose criteria rather than merely inherit them.

The vote for $N = 1$ is therefore not a biological claim that one state must always dominate. It is a context-specific institutional selection.

17 Psychology and Psychiatry: Cognitive Architecture of Dis-sent

The minority examination reveals that identical observable votes can arise from different cognitive processes.

Let

$$V_i = g_i(I_i, M_i, H_i), \quad (84)$$

where

- I_i is information;
- M_i is motivation;
- H_i is heuristic or cognitive structure.

Then

$$V_i = V_j \tag{85}$$

does not imply

$$I_i = I_j, \quad M_i = M_j, \quad H_i = H_j. \tag{86}$$

This explains why a ballot count is not a complete theory of disagreement. Psychologically, the three dissenters correspond to different concerns:

$$\text{Kant} \rightarrow \text{procedural justification}, \tag{87}$$

$$\text{Schopenhauer} \rightarrow \text{fear of domination and false unity}, \tag{88}$$

$$\text{Machiavelli} \rightarrow \text{strategic comparison and resilience}. \tag{89}$$

The institutional implication is that a Republic should avoid treating all minority votes as psychologically or normatively identical.

18 Philosophy

The deliberation contains four philosophical layers.

18.1 Plato: Unity Under Examination

Plato's preference for $N = 1$ does not establish that unity is automatically good. Instead, unity must survive dialectical examination.

18.2 Kant: Critical Justification

Kant's role is to prevent

$$\text{Preference} \rightarrow \text{Unjustified Elimination}. \tag{90}$$

His reconsideration demonstrates that critical reason can revise its own position when the relevant distinction is fully specified.

18.3 Schopenhauer: Suspicion of Unity

Schopenhauer introduces the warning

$$\boxed{\text{Unity} \neq \text{Liberation}.} \tag{91}$$

His eventual revision demonstrates the reciprocal proposition:

$$\boxed{\text{Plurality} \neq \text{Virtue}.} \tag{92}$$

18.4 Machiavelli: Examination of the Game

Machiavelli asks not merely

$$\text{Who rules?} \tag{93}$$

but

$$\boxed{\text{Who determines the game in which rulers rule?}} \quad (94)$$

This question converts political theory into the examination of information architectures, feasible sets, selection mechanisms, and higher-order control.

19 Religion and Moral Philosophy

The paper does not assert that a particular religion mechanically determines the outcome. Instead, religious and moral traditions can contribute a general principle:

$$\boxed{\text{Power over the future must remain morally examinable.}} \quad (95)$$

A risk-free rate is not merely a number when it becomes part of a permanent institutional architecture. It may affect

- intergenerational distribution;
- debt structures;
- incentives;
- accumulation;
- social hierarchy;
- conceptions of obligation.

Therefore, the economic decision is also a moral decision.

20 Political Science and Constitutional Theory

The central political principle of the paper is

$$\boxed{\text{Recognition} \neq \text{Surrender.}} \quad (96)$$

A Republic may recognize exceptional merit without transferring political sovereignty to the person recognized.

Likewise,

$$\boxed{\text{Dissent} \neq \text{Disloyalty.}} \quad (97)$$

The minority examination institutionalizes this distinction.

The Republic becomes stronger when it can perform the following operation:

$$\text{Majority Decision} + \text{Minority Examination} \rightarrow \text{Higher-Order Legitimacy.} \quad (98)$$

21 Geopolitics, Diplomacy, and International Relations

The theory also has an international interpretation.

A Republic that evaluates foreign ideas according to origin alone may suffer from epistemic protectionism.

Let a proposition be

$$P = (O, Q), \quad (99)$$

where

- O denotes origin;
- Q denotes argumentative or empirical quality.

A rational examination principle requires

$$\frac{\partial \text{Evaluation}}{\partial Q} > 0. \quad (100)$$

The origin variable may provide context but should not replace substantive evaluation.
Thus,

$$\boxed{\text{Foreign Origin} \neq \text{Epistemic Inferiority.}} \quad (101)$$

This principle supports diplomatic and international intellectual exchange without requiring political submission.

22 Warfare Economics and Strategic Choice

Warfare economics studies the allocation of resources under existential threat and strategic competition.

A Republic's discounting architecture affects strategic patience.

Let

$$U_t = B_t - C_t + \beta U_{t+1}, \quad (102)$$

where β is an intertemporal valuation parameter.

The institutional choice of a discounting mechanism therefore affects the relative weight placed on

- immediate defence;
- long-run resilience;
- capital accumulation;
- reconstruction;
- deterrence;
- intergenerational burdens.

The political lesson is:

$$\boxed{\text{Temporal architecture is strategic architecture.}} \quad (103)$$

A Republic that examines its military strategy but not its intertemporal valuation may leave an important strategic parameter unexamined.

23 Statistics and Data Science

The distinction between the initial vote and the final vote can be represented as an information update.

Let

$$\mathcal{I}^{(0)} \tag{104}$$

be the initial information set and let

$$\mathcal{F} \tag{105}$$

be additional relevant information.

The completed information set is

$$\mathcal{I}^{(1)} = \mathcal{I}^{(0)} \cup \mathcal{F}. \tag{106}$$

A Bayesian representation would write

$$P(N = 1 \mid \mathcal{I}^{(1)}) \propto P(\mathcal{F} \mid N = 1)P(N = 1 \mid \mathcal{I}^{(0)}). \tag{107}$$

The philosophical deliberation is not literally a statistical experiment, but the notation illustrates a general point:

Rational judgment may change when information changes.

(108)

24 Computer Science and Algorithms

The deliberative process can be formalized as an algorithm.

24.1 Pseudo-code

ALGORITHM DeliberativeConstitutionalChoice

INPUT:

Alternatives $E = \{E_1, E_2, \dots, E_k\}$
 Voters $V = \{v_1, v_2, \dots, v_n\}$
 Initial information sets I_i

STEP 1:

Collect initial votes.

STEP 2:

Compute provisional outcome.

STEP 3:

Identify dissenting set D .

STEP 4:

FOR each voter i in D :
 Record explicit reason R_i .
 Record relevant assumptions A_i .
 Identify missing information F_i .

STEP 5:

Construct completed information set:
I_i' = I_i union F

STEP 6:

Permit independent reconsideration.

STEP 7:

Collect revised votes.

STEP 8:

IF all revised votes agree:
Record deliberative unanimity.
ELSE:
Record remaining disagreement and reasons.

OUTPUT:

Final decision, vote structure, and reason structure.

The computational complexity of the basic vote count is

$$O(n). \quad (109)$$

If every dissenting reason must be compared with every other reason, a naive pairwise procedure may require

$$O(d^2), \quad (110)$$

where

$$d = |\mathcal{D}|. \quad (111)$$

More sophisticated clustering methods may reduce practical complexity when reason structures are represented as vectors.

25 Artificial Intelligence and Machine Learning

An AI system participating in institutional decision support should not merely predict the majority outcome.

Let the input be

$$X = (\text{votes}, \text{reasons}, \text{information}, \text{constraints}). \quad (112)$$

A naive classifier computes

$$\hat{y} = f(X) \quad (113)$$

and returns the majority decision.

A deliberative system instead constructs latent disagreement clusters:

$$Z_i = h(R_i, I_i). \quad (114)$$

The AI objective is therefore not merely

$$\min_{\theta} \mathcal{L}_{\text{prediction}}, \quad (115)$$

but potentially

$$\min_{\theta} [\mathcal{L}_{\text{prediction}} + \lambda_1 \mathcal{L}_{\text{reason}} + \lambda_2 \mathcal{L}_{\text{information}} + \lambda_3 \mathcal{L}_{\text{fairness}}]. \quad (116)$$

An appropriate constitutional AI should therefore identify:

- whether dissenters disagree for the same reason;
- whether the disagreement results from missing information;
- whether alternatives are being evaluated symmetrically;
- whether a majority is suppressing a minority rather than answering it.

26 TikZ Representation of the Political Architecture

Figure 1 represents the Political Mastery problem.

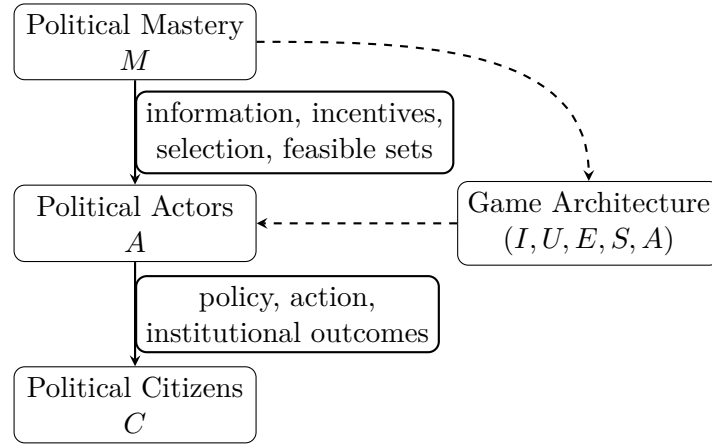


Figure 1: Political action and higher-order game architecture

The figure emphasizes that mastery may operate not only through commands but through the architecture of information, incentives, expectations, institutions, and feasible actions.

27 TikZ Representation of the Deliberative Process

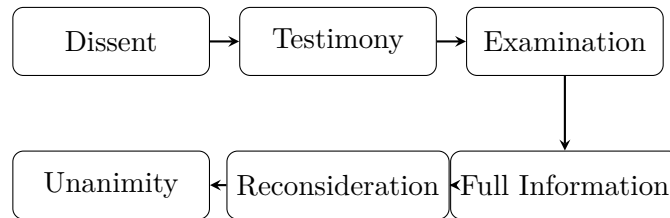


Figure 2: The deliberative route from disagreement to unanimous judgment

28 Decision Tree for N

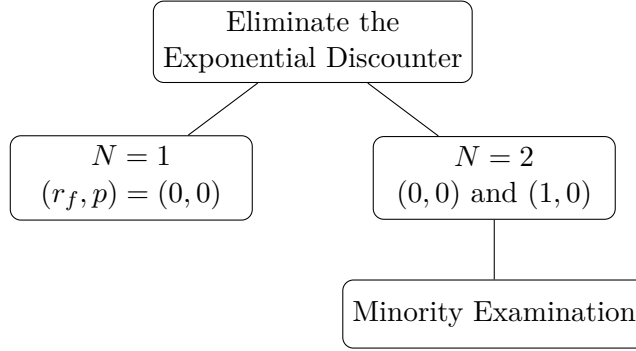


Figure 3: The constitutional-economic decision tree

29 A Simple Computational Simulation

A simulation can represent independent voter reconsideration.

Let

$$q_i = P(v_i = 1 \mid \mathcal{I}_i). \quad (117)$$

Initially, three voters possess

$$q_i < \frac{1}{2}. \quad (118)$$

After information update,

$$q'_i = P(v_i = 1 \mid \mathcal{I}_i \cup \mathcal{F}). \quad (119)$$

If

$$q'_i > \frac{1}{2} \quad (120)$$

for all three dissenters, then the revised vote becomes unanimous.
The procedure can be represented by the following pseudo-code.

```

FUNCTION UpdateVote(voter, information):
    posterior = Evaluate(voter.prior,
                        information,
                        voter.normative_constraints)

    IF posterior supports N = 1:
        RETURN 1
    ELSE:
        RETURN 2
END FUNCTION
  
```

The important property is independence. The algorithm should not impose

$$v_i^{(1)} = v_j^{(1)} \quad (121)$$

by construction.

Rather, unanimity must emerge from separate evaluations.

30 An Interdisciplinary Synthesis

The same decision can now be interpreted through multiple disciplines.

Table 4: Interdisciplinary interpretation of the vote

Field	Interpretation
Mathematics	Choice among admissible solution structures
Economics	Selection of the number of post-elimination economies
Finance	Choice involving r_f , p , and discounting architecture
Physics	Selection of a state or stable phase under constraints
Chemistry	Selection among alternative elimination pathways and resulting states
Biology	Institutional selection under a multidimensional fitness criterion
Psychology	Different motivations can produce identical observable votes
Psychiatry	Decision processes must distinguish belief, motivation, and information structures
Philosophy	Unity, plurality, reason, critique, and moral justification
Religion	Ethical examination of temporal claims and institutional obligations
Political Science	Majority rule supplemented by minority examination
Political Mastery	Examination of the mechanism structuring the game itself
Geopolitics	Evaluation of ideas independent of foreign origin
Diplomacy	Recognition without political surrender
International Relations	Institutional openness to external knowledge under reciprocal scrutiny
Welfare Economics	Comparison of benefits, costs, risks, and moral externalities
Warfare Economics	Temporal valuation as a strategic resource-allocation parameter
Statistics	Information updating and revised inference
Data Science	Representation and clustering of heterogeneous reasons for dissent
Computer Science	Algorithmic deliberation and independent vote revision
AI/ML	Reason-sensitive decision support rather than majority prediction alone
Game Theory	Choice-set formation and higher-order control of strategic environments
Constitutional Theory	Legitimacy through examination rather than mere numerical victory

31 General Proposition on Republican Self-Examination

Proposition 3. *A Republic is not fully self-governing merely because citizens choose among available alternatives.*

Proof. Let citizens choose

$$a_i \in A_i. \tag{122}$$

If another mechanism determines A_i , then the citizen may exercise local choice without controlling the formation of the feasible set.

Therefore,

$$\text{Choice within } A_i \not\Rightarrow \text{Control over } A_i. \quad (123)$$

If information, incentives, institutions, and selection mechanisms are also externally or opaquely determined, then formal choice alone does not guarantee higher-order autonomy.

Thus a self-governing Republic must be able to examine

$$\boxed{I_i, \quad U_i, \quad E_i, \quad S_i, \quad A_i.} \quad (124)$$

The conclusion follows. □

32 The Economic Constitutional Record

The monumental vote may therefore be recorded as follows:

The Republic considered the alternatives $N = 1$ and $N = 2$. The initial vote was 4–3 in favour of $N = 1$. The minority was not silenced. Its members were asked to state their specific reasons. Those reasons were found to be heterogeneous. The relevant information concerning the economic structures was made explicit. Each dissenting voter was permitted to reconsider independently. The final vote was 7–0 in favour of $N = 1$.

The constitutional significance is therefore

$$\boxed{\text{Unanimity was not imposed at the beginning.}} \quad (125)$$

Instead,

$$\boxed{\text{Unanimity emerged after dissent was examined.}} \quad (126)$$

The final economic selection is

$$\boxed{N = 1, \quad r_f = 0, \quad p = 0.} \quad (127)$$

33 Discussion

Several broader conclusions follow.

First, a theorem can produce multiple mathematically admissible solutions without requiring every solution to become an institutional equilibrium.

Second, majority voting is incomplete when the reasons for minority dissent remain unknown.

Third, the same ballot may conceal radically different motivations.

Fourth, political autonomy requires examination of the structures that produce feasible choices.

Fifth, a discounting mechanism is not necessarily a neutral technical parameter. It may be part of the constitutional architecture through which a society values the future.

Sixth, moral evaluation can legitimately enter institutional economic choice without replacing mathematical analysis. The correct relation is

$$\boxed{\text{Mathematics} + \text{Economics} + \text{Institutional Analysis} + \text{Moral Examination.}} \quad (128)$$

Seventh, unanimity is most legitimate when it is the consequence of independent reconsideration rather than prior coercion.

34 Conclusion

The monumental vote on N began as a narrow choice between one and two economies.

It became a broader inquiry into

- discount factors;
- risk-free rates;
- risk premia;
- institutional elimination;
- political mastery;
- feasible-set formation;
- minority rights;
- moral evaluation;
- welfare;
- strategic resilience;
- information updating;
- algorithmic deliberation;
- and republican self-examination.

The decisive methodological principle is:

$$\boxed{\text{Do not merely count dissent. Examine it.}} \quad (129)$$

The decisive political principle is:

$$\boxed{\text{Do not merely examine rulers. Examine the game that produces rulers.}} \quad (130)$$

The decisive economic result, within the theorem architecture and deliberative framework considered in this paper, is:

$$\boxed{N = 1, \quad r_f = 0, \quad p = 0.} \quad (131)$$

The decisive constitutional result is:

$$\boxed{\text{Dissent} \rightarrow \text{Examination} \rightarrow \text{Full Information} \rightarrow \text{Reconsideration} \rightarrow \text{Unanimity.}} \quad (132)$$

Thus the vote is recorded not simply as a numerical victory.

It is recorded as a procedure of republican self-examination.

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Glossary

Admissible Economy

An economy satisfying the mathematical conditions of the relevant theoretical architecture.

Deliberative Unanimity

Unanimity emerging after independent examination and reconsideration rather than through suppression of dissent.

Discount Factor

A quantity converting future value into present value. A simple example is $D = 1/(1+r)$.

Exponential Discounter

A mechanism in which future value is discounted exponentially, commonly of the form $D_t = (1+r)^{-t}$.

Feasible Set

The set A_i of actions available to an actor.

Full Information Set

The information available after relevant omissions or ambiguities have been incorporated into deliberation.

Minority Examination

A procedure that identifies and evaluates the specific reasons and information structures underlying dissenting votes.

Political Actor

An individual, organization, or institution making visible political choices.

Political Citizen

A member of the political community affected by and participating in the political system.

Political Mastery

Higher-order influence or control over the information, incentives, institutions, selection mechanisms, or feasible sets within which political actors operate.

Risk-Free Rate

The rate r_f assigned to a risk-free return in the theoretical framework.

Risk Premium

The quantity p representing compensation for risk in the theoretical framework.

Republican Self-Examination

The capacity of a Republic to examine not only visible rulers but also the architectures that structure political and economic choice.

Unnecessary Repetition

The retention of an additional institutional structure without a demonstrated compensating benefit.

The End